

Renormalized Area of Surfaces in $\mathbb{H}^3$

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Part I

The Central Identity:

Renormalized Area as the Difference Between Bending Energies

Problem Setting

Consider the upper half-space model for the hyperbolic space $\mathbb{H}^3$:

$$\mathbb{H}^3 = \{(x, y, z) \in \mathbb{R}^3 \,|\, z > 0\},$$

endowed with the metric

$$g = \frac{1}{z^2}\left(dx^2 + dy^2 + dz^2\right).$$

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Let M be a smooth surface properly embedded in $\mathbb{H}^3$ and assume that M extends smoothly to the ideal boundary $\partial_\infty \mathbb{H}^3 = \{z = 0\}$ approaching it at a right angle. Then, the area of M,

$$\mathcal{A}(M) = \int_M 1\, dA\,,$$

is infinite, but it possesses a well-defined Hadamard regularization, known as the renormalized area $\mathcal{A}_R(M)$.

Renormalized Area

The renormalized area $\mathcal{A}_R$ of M is the constant term in the expansion of the area of the truncated surfaces $M_\epsilon = M \cap \{z \geq \epsilon\}$:

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- In 1999, C. Graham and E. Witten introduced $\mathcal{A}_R$ for minimal submanifolds, motivated by the AdS/CFT correspondence.
- In 2010, S. Alexakis and R. Mazzeo further analyzed $\mathcal{A}_R$ for general surfaces, obtaining

$$\mathcal{A}_R(M) = -2\pi\chi(M) - \frac{1}{2}\int_M |\mathring{B}|^2\, dA + \int_M H^2\, dA\,.$$

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After the conformal change of metric $\overline{g} = z^2 g$, M can be regarded as a surface in $\mathbb{R}^3_+$ with boundary ∂M in the plane $\{z = 0\}$ and meeting this plane orthogonally.

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Willmore Energy **(Alexakis & Mazzeo, 2010)**

If M is minimal in $\mathbb{H}^3$, then

$$\mathcal{A}_R(M) = -\mathcal{W}(M) = -\int_M \overline{H}^2 \, d\overline{A}\,.$$

(The Euclidean objects are denoted with overbars.)

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(The Euclidean objects are denoted with overbars.)

- Alexakis-Mazzeo: "On the other hand, it does not seem to be the case that the extended functional [$\mathcal{A}_R$ on non-minimal surfaces] has any simple connection with the Willmore functional."

Difference of Bending Energies

Theorem **(Palmer & P., 2026), (P., 2026)**

The renormalized area of M is the difference between the hyperbolic and Euclidean bending energies:

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- Rigidity **(P. & Tyrrell, 2026-2)**: $\mathcal{A}_R(M) = -\mathcal{W}(M)$ if and only if M is minimal in $\mathbb{H}^3$.
- Natural question: $\mathcal{A}_R(M) = \lambda\mathcal{W}(M)$?

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$$\begin{aligned}
\overline{\nabla} \cdot \overline{\nabla}(\log z) &= 2\overline{H}\,\frac{\overline{\xi}_3}{z} + \frac{\overline{\xi}_3^2}{z^2} - \frac{1}{z^2} \\
&= \left(\overline{H} + \frac{\overline{\xi}_3}{z}\right)^2 - \overline{H}^2 - \frac{1}{z^2} \\
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The result then follows by integrating over M_ϵ, applying the Divergence Theorem, and letting $\epsilon \to 0^+$.

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- This latter proof can be extended to higher dimensions using Chen's conformally invariant quantity:

$$\left(H^2 - R\right) g\,.$$

Part II

Geometric Applications of the Central Identity:

The Proportionality Relation and Consequences

Proportionality Relation

The meaningful problem is to identify geometric surface equations for which there exists a universal constant $\lambda \in \mathbb{R}$ such that

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Theorem (P. & Tyrrell, 2026-2)

Let M be a topological annulus satisfying

$$a_1 H^2 + a_2 R + a_3 \left(H - \nu\right)^2 = 0\,,$$

for constants $a_1, a_2, a_3 \in \mathbb{R}$, not all zero. Then, $a_1 + a_2 \neq 0$ and

$$\mathcal{A}_R(M) = -\frac{a_1 + a_3}{a_1 + a_2}\,\mathcal{W}(M)\,.$$

(If $a_2 = 0$, the topological restriction is not needed.)

Linear Weingarten Annuli

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Corollary (P. & Tyrrell, 2026-2)

Let M be a topological annulus satisfying $\kappa_2 = \mu\kappa_1$ for $\mu \neq 0, 1$. Then,

$$\mathcal{A}_R(M) = \frac{4\mu}{(1-\mu)^2}\,\mathcal{W}(M)\,.$$

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- **(P., 2018), (Arroyo, Garay & P., 2018), (López & P., 2020)** The generating curve of M is critical for

$$\mathbf{\Theta}_p(\gamma) = \int_\gamma \kappa^p\, ds\,, \qquad p = \frac{1}{1-\mu}\,.$$

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- **(P. & Tyrrell, 2026-2)** For every $\mu < 0$ and $c > 0$, M is a topological annulus that approaches $\partial_\infty \mathbb{H}^3$ at a right angle.

Explicit Computations of $\mathcal{A}_R$

We have a biparametric family of admissible rotational linear Weingarten surfaces $M \cong M_{\mu,c}$ for $\mu < 0$ and $c > 0$.

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The renormalized area of $M_{\mu,c}$ is the convergent improper integral

$$\mathcal{A}_R(M_{\mu,c}) = \frac{-4\mu^2\pi}{c} \int_0^{\alpha} \frac{x^{1-\mu}}{\sqrt{Q_{\mu,c}(x)}}\, dx\,,$$

where $Q_{\mu,c}(x) = -\mu^2 x^{2(1-\mu)} + c^2 x^{-2\mu} + 1$ and $\alpha > 0$ is its only positive solution of $Q_{\mu,c}(x) = 0$.

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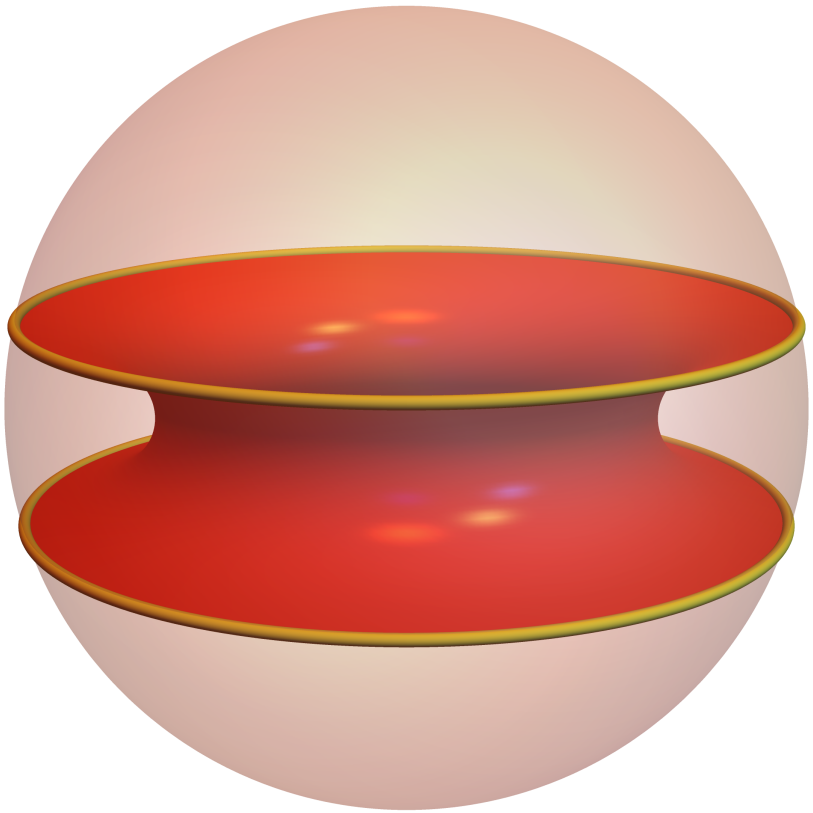
- $\mathcal{A}_R(M_{\mu,c})$ takes every value in $(-\infty, 0)$.

Catenoids in $\mathbb{H}^3$

When $\mu = -1$, the annuli $M_{-1,c}$ are spherical catenoids in $\mathbb{H}^3$.

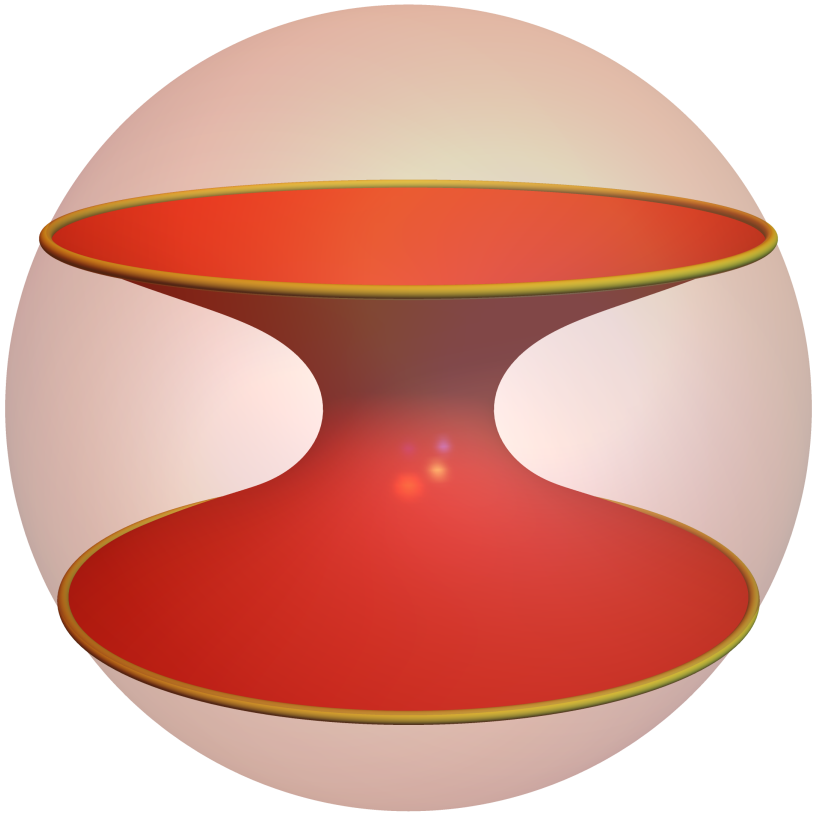
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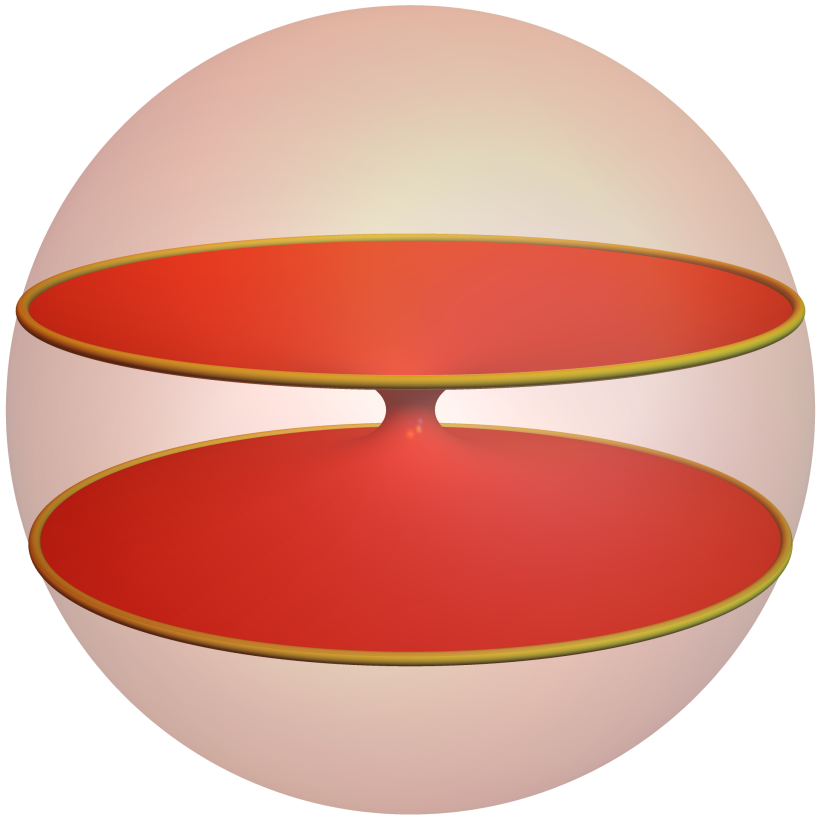
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Renormalized Area of Catenoids

Proposition (P. & Tyrrell, 2026-1)

The renormalized area of a spherical catenoid $M_{-1,c}$ in $\mathbb{H}^3$ is

$$\mathcal{A}_R(M_{-1,c}) = \frac{4\pi}{\sqrt{2\zeta^2 - 1}} \left((1-\zeta^2)K(\zeta) - E(\zeta) \right),$$

where

$$\zeta = \sqrt{\frac{c^2 + \sqrt{c^4+4}}{2\sqrt{c^4+4}}} \in (\sqrt{2}/2, 1).$$

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$$\lim_{c \to 0^+} \mathcal{A}_R(M_{-1,c}) = -\infty\,, \qquad \lim_{c \to \infty} \mathcal{A}_R(M_{-1,c}) = 2\mathcal{A}_R(\mathbb{H}^2) = -4\pi\,.$$

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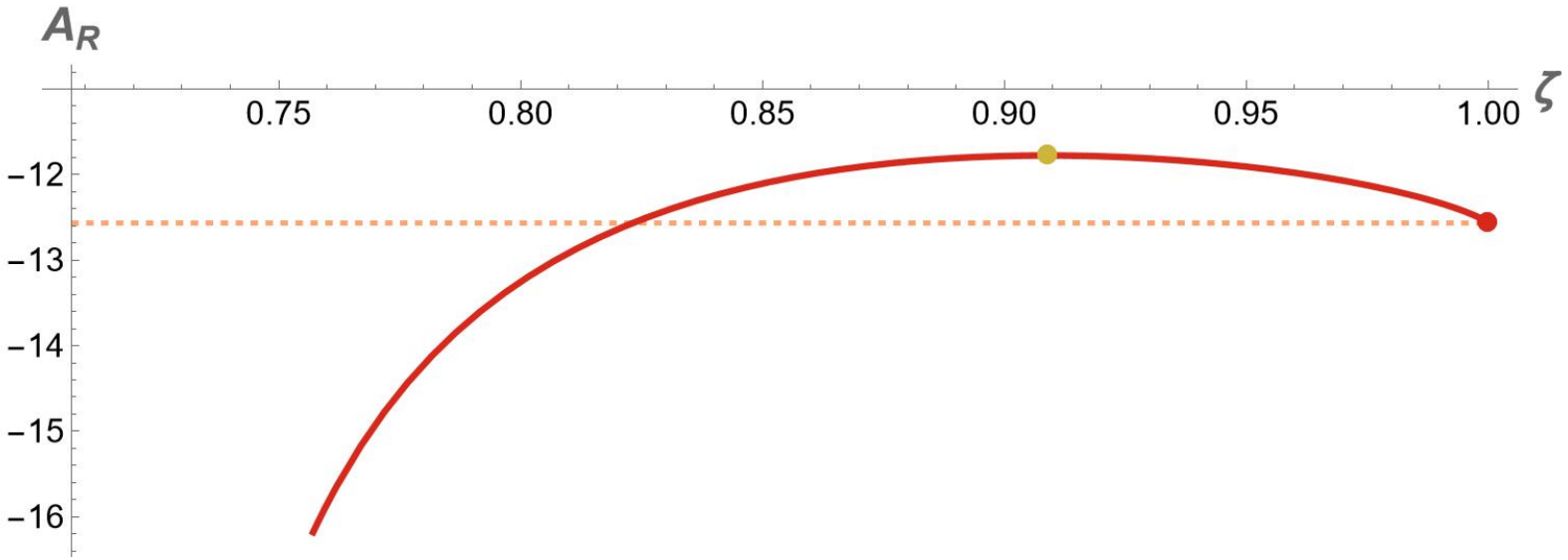


FIGURE: Graph of $\mathcal{A}_R(M_{-1,c})$, $\zeta \in (\sqrt{2}/2, 1)$.

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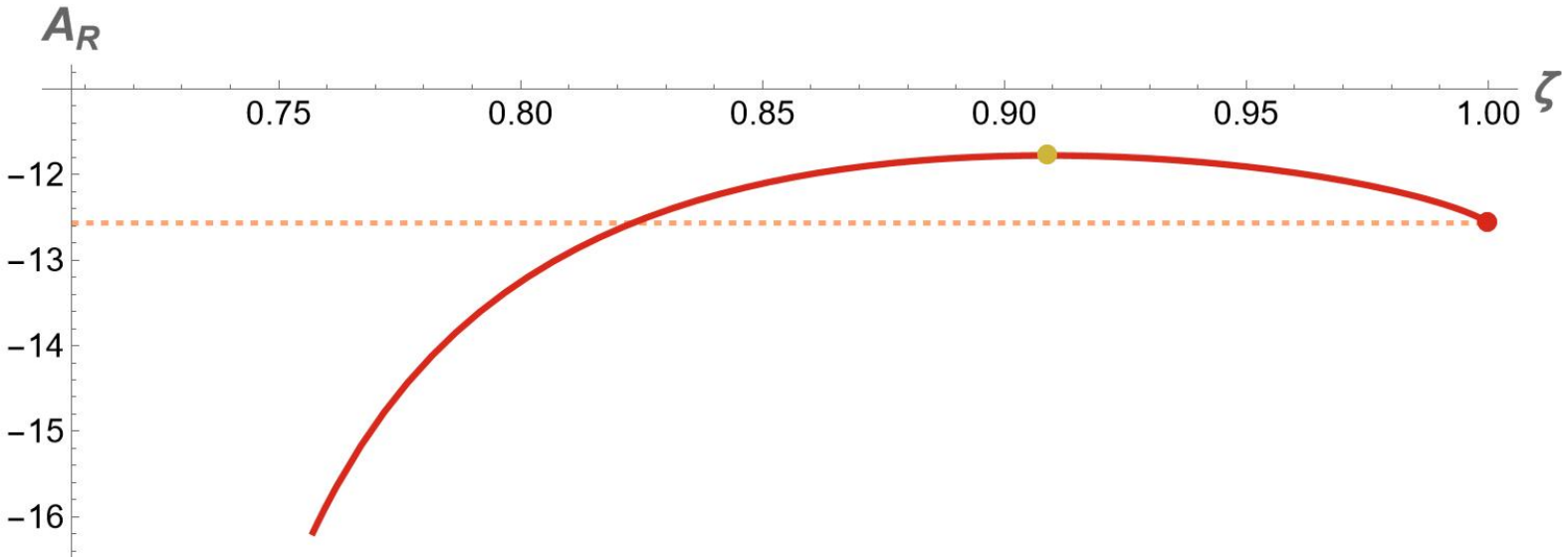


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Corollary **(P. & Tyrrell, 2026-1)**

The maximum of $\mathcal{A}_R(M_{-1,c})$ is attained at the catenoid with maximum height.

Explicit Computations of $\mathcal{W}$

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Theorem (P. & Tyrrell, 2026-2)

The Willmore energy of $M_{\mu,c}$, regarded as a surface in $\mathbb{R}^3_+$, is the convergent improper integral

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Moreover, $\mathcal{W}(M_{\mu,c})$ takes every value in (π^2, ∞).

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Theorem (P. & Tyrrell, 2026-2)

Let $\mu < 0$ and $c = -\mu/\sqrt{2}$. As $\mu \to -\infty$, the doubles of $M_{\mu,c}$ converge in $\mathcal{W}^{2,p}$, for every $p \in [1, \infty)$, to the rotational Clifford torus.

Convergence to the Clifford Torus



FIGURE: Double of $M_{\mu,c}$ with $c = -\mu/\sqrt{2}$ and $\mu = -0.6$.

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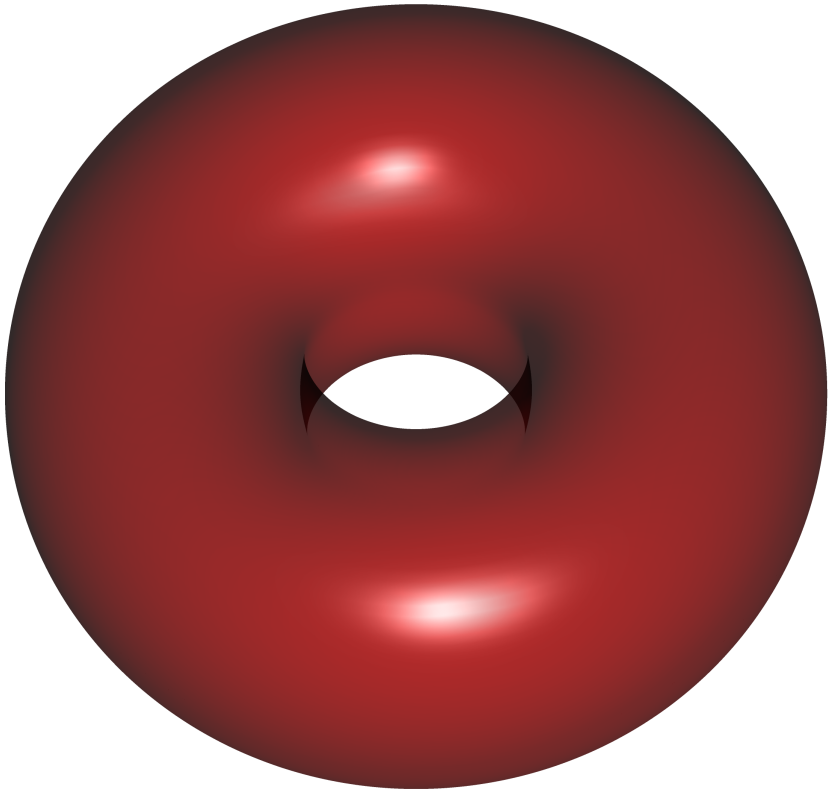


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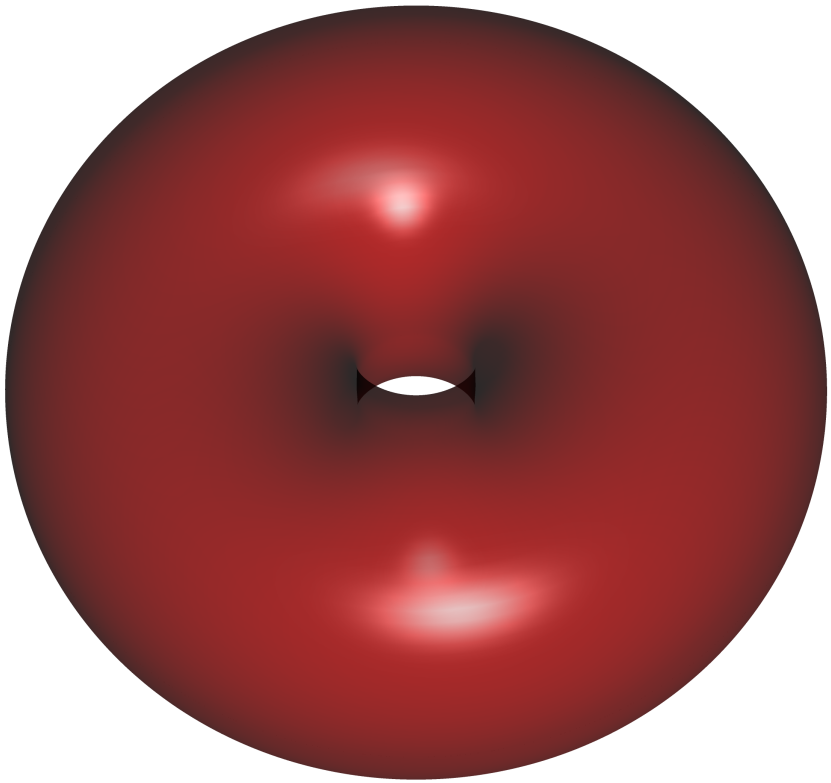


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Part III

Variational Consequences of the Central Identity

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- Using compactly supported normal variations, we deduce that $H = 0$ in the interior.
- Then, differentiation of the central identity gives

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Proposition **(Palmer & P., 2026)**

Equilibrium surfaces for $\mathcal{A}_R$ are minimal surfaces in $\mathbb{H}^3$ satisfying $\partial_{\overline{n}}\overline{H} = 0$ along the boundary.

Previous Rigidity Result

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- They used the asymptotic expansion of the graphing function $u(s, z)$.
- Their boundary condition is the vanishing of the coefficient u_3 in this expansion.
- Using an adapted maximum principle at the ideal boundary, they proved:

Theorem (Alexakis & Mazzeo, 2010)

Equilibrium surfaces with connected asymptotic boundary for $\mathcal{A}_R$ are totally geodesic discs of $\mathbb{H}^3$.

Complementary Rigidity Result

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- The differential Φ can be holomorphically reflected to a closed doubled surface of genus $\mathfrak{b} - 1$.
- The Four-Vertex Theorem guarantees that on each boundary component the extended Φ has at least four zeros.
- A nonzero holomorphic quadratic differential on the double surface has $4(\mathfrak{b} - 1) - 4 = 4\mathfrak{b} - 8$ zeros.

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Sketch of the proof. Let M be a genus zero minimal surface in $\mathbb{H}^3$ with $\mathfrak{b}$ boundary components along which $\partial_{\overline{n}} \overline{H} = 0$.

- Consider the hyperbolic Hopf differential $\Phi = \overline{\Phi}/z$.
- The differential Φ can be holomorphically reflected to a closed doubled surface of genus $\mathfrak{b} - 1$.
- The Four-Vertex Theorem guarantees that on each boundary component the extended Φ has at least four zeros.
- A nonzero holomorphic quadratic differential on the double surface has $4(\mathfrak{b} - 1) - 4 = 4\mathfrak{b} - 8$ zeros. Hence, Φ is identically zero, and so is $\overline{\Phi}$.

THE END

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Thank You!